

Tyler Katz

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EDUCATION

Syracuse University, School of Information Studies

Master of Science in Applied Human-Centered Artificial Intelligence

Syracuse, NY

May 2027

Syracuse University, School of Information Studies

Bachelor of Science in Applied Data Analytics

Syracuse, NY

May 2026

- GPA: 3.9/4.0; Dean's List (6 semesters)

WORK & LEADERSHIP EXPERIENCE

Comcast

New York, NY (Remote)

Data Science Intern

June - August 2026

- Automated quarterly network device inventory reconciliation by building an LLM agent and MCP integration, cutting the process from hundreds of hours to 5 minutes and reducing token costs 30–40%.
- Engineered a FastAPI pipeline across 14 device platforms to auto-generate and upload CMDB templates, improving data consistency for ~5,000 network devices.
- Applied the agent to generate structured discrepancy reports across 14 platforms, cutting total reporting time from hundreds of hours to 2–3 hours.

United AI

Syracuse, NY

Engineer Lead

September - December 2025

- Led 10 student AI/ML teams, meeting weekly to unblock issues and guide projects to production.
- Served as technical lead, advising on Python and model training/deployment, and reviewing designs for code quality.
- Managed a 10-project GitHub org (repos, CI/CD) and lectured 200+ members on Git, Claude Code, and new AI tools.

Syracuse University

Syracuse, NY

Undergraduate Data Analytics Research Assistant

January - December 2025

- Engineer clean, integrated datasets from raw census files, geospatial shapefiles, and web-scraped sources using R, streamlining data preparation workflows and improving analysis efficiency for the project
- Design and implement data visualizations, geospatial maps, and statistical tables that were directly incorporated into multiple chapters of *Machine [Gun] Politics*, enhancing the clarity and persuasiveness of research findings
- Translate complex, multi-source datasets into actionable insights through advanced data wrangling, exploratory analysis, and visualization techniques, driving progress toward book completion and supporting evidence-based conclusions

PROJECTS

Fine-tuned LLM Microservice | Python, QLoRA, Docker, FastAPI, AWS | [GitHub Repository](#)

August 2026

- Fine-tuned Qwen2.5-3B-Instruct via QLoRA (4-bit) for structured extraction of skills, tech stack, seniority, and comp from job postings, more than 8x'ing tech-stack F1 and lifting comp accuracy to 97% over the zero-shot baseline.
- Diagnosed a regressed fine-tune via raw-generation inspection, traced it to noisy data, and hit 100% valid-JSON on retrain.
- Containerized the fine-tuned model into a FastAPI service, deployed to AWS EC2 (Graviton/ARM) behind a public endpoint, and wired a GitHub Actions CI/CD pipeline to test, build, and push on every commit, plus latency-logging middleware.

Agentic Rag System | Python, LangGraph, Docker, FastAPI, Streamlit | [GitHub Repository](#)

July 2026

- Designed a 4-node LangGraph agent that decides when to retrieve, reformulates weak queries, and cites sources.
- Built an eval harness (hit-rate + LLM-as-judge) showing agent beats baseline: 90% vs 85% hit-rate, 2.80 vs 2.75 faithfulness.
- Shipped a Dockerized FastAPI + Streamlit RAG service answering questions over 100 job postings and 250 papers.

Inventory Reconciliation Agent - Comcast (Internal Tool) | Python, LLM Agent, MCP Server

June - July 2026

- Built an LLM reconciliation agent on a new in-house platform, one of the first use cases to demonstrate its effectiveness.
- Redesigning the agent's architecture from an executable skill (script run and summarized in-platform) to an instructional skill (script run externally, output uploaded with structured formatting instructions), cutting token usage 30–40%.
- Developed a custom MCP integration enabling the agent to autonomously remediate its own reconciliation recommendations.

TECHNICAL SKILLS & CERTIFICATIONS

Programming Languages: Python | JavaScript | R | SQL | HTML | CSS

Frameworks & Tools: FastAPI | PyTorch | AWS | Docker | Git | MCP | Tableau | Power BI

Software: Microsoft Office | MS SQL Server Management Studio

Certifications: Microsoft Office Specialist (MOS): Excel Associate | Google Analytics | Advanced Google Analytics